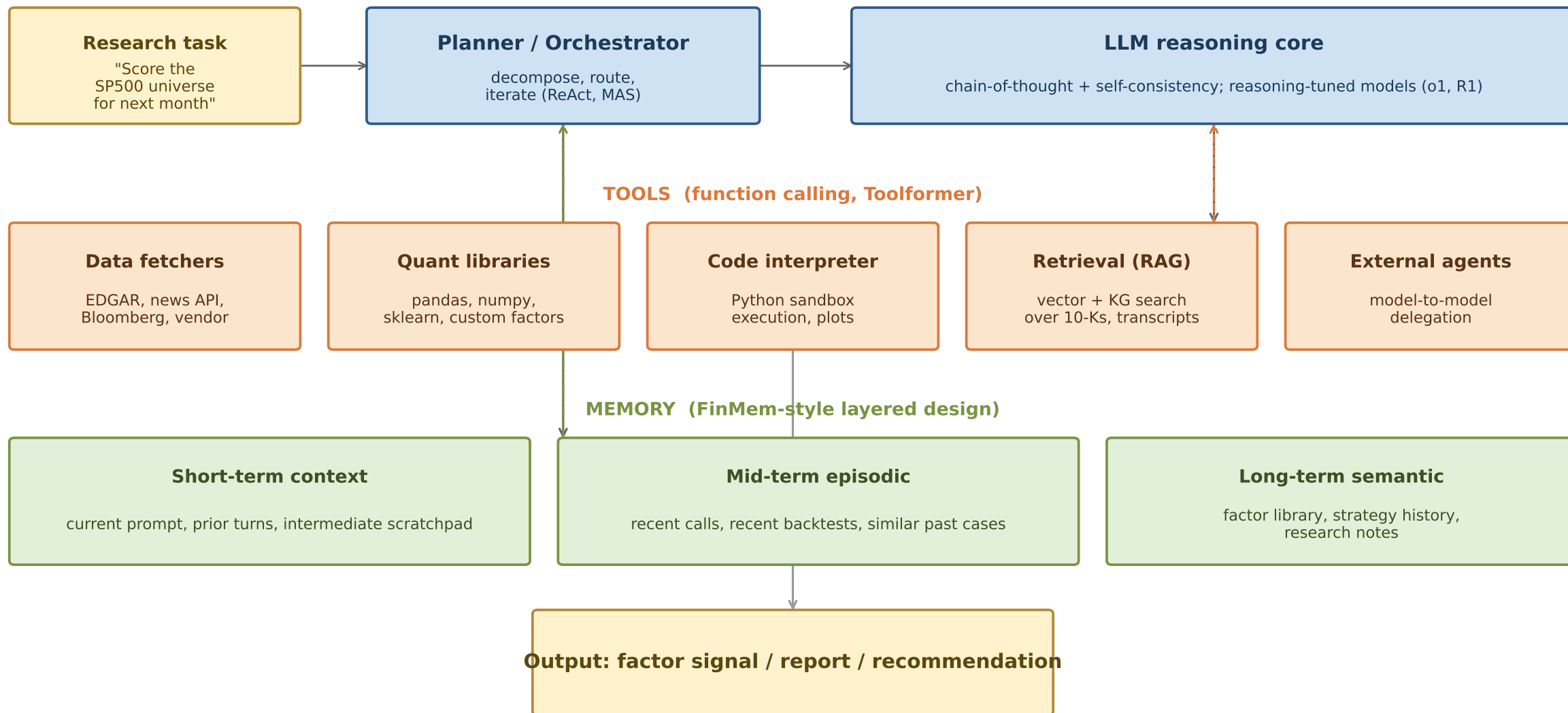


An LLM-agent architecture for financial research

Reasoning core, tool layer, memory, and orchestration: the four ingredients of an agentic pipeline



Reading order: task → planner → reasoning core (with iterative tool calls and memory reads) → output. Single-prompt LLMs use only the reasoning core; agents add the orange tool layer and the green memory layer. FinMem (Yu et al., 2024) and FinRobot (Yang et al., 2024) instantiate this template for trading and analyst workflows; TradingGPT (Li et al., 2023) and Du et al. (2023) extend it to multi-agent debate.

FIGURE 23.2: LLM-agent architecture